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COMPUTER ARCHİTECTURE COE-301 PROJECT

Özgül Rana ÇOKKAYIŞLI
220611017

THE MEMORY WALL CHALLENGE

Content:

Cache memory represents one of the most critical design decisions in modern computer architecture. The challenge lies not in building the largest or most complex cache, but rather in designing the right cache for the specific workload at hand.

A cache configuration that delivers exceptional performance for one application may prove entirely wasteful for another, consuming valuable silicon area and power without providing corresponding benefits.

Research Question:

Can artificial intelligence effectively design optimal cache configurations, or does human architectural expertise provide measurable advantages through systematic experimentation? This study evaluates both approaches through analysis of five distinct memory access patterns using the UW CSE 351 Cache Simulator.

RESEARCH METHODOLOGY

AI METHODOLOGY:

- Pattern recognition
- Theoretical modeling
- Heuristic application
- Rapid prediction
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HUMAN METHODOLOGY:

- Hypothesis formation
- Controlled experimentation
- Empirical data collection
- Statistical validation
- Cost-benefit analysis

Experimental Scope:

- 5 distinct memory traces analyzed
- 80+ configuration combinations tested
- 4 parameters systematically varied (Cache Size, Block Size, Associativity, Replacement Policy)
- UW CSE 351 Cache Simulator employed

TRACE 1 ANALYSIS

Pattern: 16, 24, 128, 16, 24, 128, 16, 24, 128, 16, 24, 128, 16, 24, 128, 16, 24, 128

Manual Analysis:

Trace 1 presents a textbook example of a small working set with intensive temporal reuse. The entire trace consists of exactly three memory addresses (16, 24, and 128) accessed in an unwavering cyclic pattern that repeats six times, generating 18 total memory operations.

Locality Hypothesis:

- Temporal Locality: Exceptional - 500% reuse ratio (each address used 6 times)
- Spatial Locality: Selective - addresses 16 and 24 are 8 bytes apart; address 128 is isolated 104 bytes away

Configuration	Cache Size	Block Size	Associativity	Hits	Misses	Hit Rate
Experiment 1	64B	8B	Direct-Mapped	15	3	83.33%
Experiment 2	64B	16B	Direct-Mapped	16	2	88.89%
Experiment 3	64B	16B	2-Way	16	2	88.89%

TRACE 1 - AI vs HUMAN COMPARISON

AI Recommendation:

Feature	Setting	Reasoning
Cache Size	64 Bytes	Large enough to hold working set
Block Size	16 Bytes	Captures addresses 16 and 24 together
Associativity	**2-Way Set Associative**	**Prevents conflict misses**
Replacement	**LRU**	**Optimal for cyclic patterns**

Human Recommendation (Based on Testing):

Feature	Setting	Empirical Finding
Cache Size	64 Bytes	Tested: 64B=128B=256B (all 83.33%)
Block Size	16 Bytes	Tested: improves from 83.33% to 88.89%
Associativity	**Direct-Mapped**	**Tested: 1-way=2-way=4-way (all 88.89%)**
Replacement	**Random**	**Tested: LRU=FIFO=Random (all 88.89%)**

Critical Finding

AI's Theoretical Assumption:

"In a direct-mapped cache, there is a risk that Block 1 and Block 8 might map to the same cache line (index). Using 2-way set associativity ensures that even if they map to the same set, they can reside in the cache simultaneously."

Human's Empirical Discovery:

The AI made a theoretical assumption about potential conflicts WITHOUT validation. When actually tested in Experiment 3, direct-mapped, 2-way, and 4-way all achieved IDENTICAL 88.89% performance with 16-byte blocks.

Why No Conflicts Occur:

1. Small working set: Only three addresses requiring at most two 16-byte blocks (16-31 and 128-143)
2. Natural set distribution: These addresses apparently map to different cache sets even in direct-mapped configuration
3. Continuous reuse: Temporal access pattern prevents harmful conflicts

TRACE 2 ANALYSIS

Sequential Pattern: 128 → 188, stride +4

Workload Characterization:

This trace demonstrates pure sequential access with 16 memory references advancing monotonically through contiguous memory. The pattern exhibits zero temporal reuse and perfect spatial locality.

Experimental Findings - Block Size Impact:

Block Size	Hits	Misses	Hit Rate
2 bytes	0	16	0%
4 bytes	0	16	0%
8 bytes	8	8	50%
16 bytes	12	4	75%

Block size is the sole determinant of performance for streaming workloads. 75% of cache parameters have zero impact.

TRACE 2 - ANALYSIS DISCREPANCY

AI Analysis Error:
The AI system incorrectly identified 12 memory accesses with maximum address 172, when the trace actually contains 16 accesses extending to address 188. This fundamental counting error invalidated subsequent performance predictions and led to recommendations for 32-byte blocks that cannot be validated experimentally in the simulator.

Human Analysis:
Systematic enumeration confirmed 16 accesses. Testing proceeded with available block sizes (2B, 4B, 8B, 16B), with empirical measurements demonstrating 75% hit rate as the achievable maximum with 16-byte blocks.

Methodological Implication:
Theoretical predictions without empirical validation can lead to design recommendations based on incorrect assumptions or unmeasurable configurations.

Result: AI provided untestable recommendations; Human provided measured, verified results.

TRACE 3 ANALYSIS

Workload Characteristics:

- Access pattern: 20, 480, 312, 800, 128, 940, 516, 624, 2, 712, 340, 998
- Mean inter-access distance: 451 bytes
- Temporal locality: Zero (no address repetition)
- Spatial locality: Zero (gaps exceed practical block sizes)

Comprehensive Experimental Results:

All configurations tested achieved 0% hit rate:

Parameter	Tested Values	Result
Cache Capacity	64B, 128B, 256B	All 0%
Block Size	2B, 4B, 8B, 16B	All 0%
Associativity	1-way, 2-way, 4-way	All 0%
Replacement Policy	LRU, FIFO, Random	All 0%

Complete absence of locality results in uniform failure across all tested configurations. Every access is a compulsory miss.

Strategic Implication:

When performance optimization proves impossible, design objective shifts to minimizing implementation cost rather than maximizing performance.

TRACE 3 - DESIGN PHILOSOPHY COMPARISON

Aspect	AI Recommendation	Human Recommendation
Cache Size	128 bytes	64 bytes
Block Size	4-8 bytes	4 bytes
Associativity	Fully associative	Direct-mapped
Replacement	LRU	Random
Design Rationale	Optimize for potential future reuse	Minimize cost when optimization fails
Relative Cost	~6.0×	1.0×
Performance	0% hit rate	0% hit rate

Analysis:
The AI approach invested significant hardware resources optimizing for theoretical scenarios not present in the workload. Fully associative configuration requires 12 parallel comparators and complex control logic.

Human analysis recognized the futility of performance optimization and instead minimized implementation overhead, selecting the simplest possible configuration that accepts 0% as inevitable.
Cost Impact: Both achieve 0% performance, but human design costs 83% less.

TRACE 4 ANALYSIS

Pattern Structure:
Four sequential clusters at 256-byte boundaries:

- Cluster 1: addresses 0-20 (6 accesses, 4-byte stride)
- Cluster 2: addresses 256-276 (6 accesses, 4-byte stride)
- Cluster 3: addresses 512-532 (6 accesses, 4-byte stride)
- Cluster 4: addresses 768-788 (6 accesses, 4-byte stride)

Locality Characteristics:

- Intra-cluster: High spatial locality (4-byte stride)
- Inter-cluster: No spatial relationship (256-byte gaps)
- Temporal: None (single pass through all clusters)

Associativity Analysis:

Configuration	Hits	Misses	Hit Rate
1-way (Direct-Mapped)	12	12	50.00%
2-way	12	12	50.00%
4-way	12	12	50.00%

Contrary to theoretical predictions of conflict misses due to 256-byte address alignment, empirical testing reveals no performance difference across associativity levels.

TRACE 4 - THEORETICAL vs EMPIRICAL RESULTS

Hypothesis Evaluation:

AI Prediction:

"Addresses 0, 256, 512, and 768 likely map to identical cache sets in direct-mapped configuration due to 256-byte spacing. 4-way set associativity is CRITICAL to prevent conflict misses."

Empirical Testing Results:

Direct-mapped, 2-way, and 4-way configurations all achieved identical 50% hit rates with 8-byte blocks.

Analysis of Discrepancy:

The single-pass access pattern with zero temporal reuse means blocks are never revisited after initial use. Therefore, even if address mapping creates set conflicts, no performance degradation occurs because evicted blocks are not subsequently referenced.

Cost Impact:

Design	Associativity	Relative Cost	Performance
AI	4-way	4.0×	75%
Human	Direct-mapped	1.0×	75%

AI recommendation requires 4× hardware complexity for associativity with zero performance benefit.

TTRACE 5 ANALYSIS

Workload Characteristics:
Pattern: Two 16-byte clusters at addresses 0-12 and 256-268, each accessed sequentially, pattern repeated three times.
This introduces temporal locality through repetition while maintaining spatial locality within clusters.

Performance Analysis:

Configuration	Block Size	Hits	Misses	Hit Rate
Direct-Mapped	8 bytes	12	12	50.00%
Direct-Mapped	16 bytes	12	12	50.00%
2-Way	8 bytes	20	4	83.33%
2-Way	16 bytes	22	2	91.67%

Temporal locality benefits are completely negated by conflict misses in direct-mapped configuration. Addresses 0 and 256 map to the same cache set, causing thrashing that defeats all temporal reuse.
Only with adequate associativity (2-way) can temporal reuse be exploited effectively.

TRACE 5 - PARAMETER INTERACTION EFFECTS

Cross-Parameter Testing Results:

Block Size	Direct-Mapped	2-Way	4-Way
8 bytes	50.00%	83.33%	83.33%
16 bytes	50.00%	91.67%	91.67%

Key Observations:

1. Block size increases provide no benefit in direct-mapped configuration - Performance remains at 50% regardless of block size due to conflict-induced thrashing
2. Once conflicts are resolved via 2-way associativity, block size optimization becomes effective - Performance improves from 83.33% to 91.67%
3. 4-way associativity provides no incremental benefit over 2-way - Both achieve identical performance for this two-cluster workload
4. Replacement policy testing showed identical results - LRU, FIFO, and Random all achieved 83.33% with 8B blocks

Implication:
Cache parameters do not operate independently. Optimization requires understanding parameter interactions and dependencies. Block size gains are worthless until conflicts are fixed.

TRACE 5

AI INTERPRETATION

The AI-based analysis interprets Trace 5 as a workload where temporal locality exists due to repetition, but is suppressed by conflict misses in a direct-mapped cache. Because the two clusters map to the same cache index, the AI correctly concludes that increasing associativity is required to enable effective reuse. Based on general cache design heuristics, the AI would prioritize associativity as the primary optimization to recover performance.

AI LIMITATION

While the AI correctly identifies conflict misses as the root cause, it does not explicitly capture the interaction between cache parameters. In particular, the AI does not recognize that increasing block size provides no benefit until conflicts are resolved. As a result, the AI reasoning remains partially theoretical and does not isolate which optimizations are conditionally effective.

TRACE 5

HUMAN EXPERIMENTAL ADVANTAGE

The human experimental analysis systematically tests parameter combinations and demonstrates that block size optimization is meaningless in a direct-mapped configuration due to conflict-induced thrashing. Only after associativity eliminates conflicts does block size become impactful, improving performance significantly. This empirical approach allows unnecessary optimizations to be eliminated and the minimal effective configuration to be identified.

AI vs HUMAN COMPARISON

Both AI and human analyses identify conflict misses as the dominant issue in Trace 5. However, the AI applies generalized heuristics, while the human approach validates assumptions through experimentation. The human analysis reveals parameter dependencies and achieves the same or better performance with a simpler and more cost-efficient cache design.

CONCLUSION

This project shows that AI-based cache recommendations are generally correct in principle but often over-engineered due to the lack of experimental validation. Human-driven analysis, supported by systematic simulation, identifies which cache parameters truly matter and eliminates unnecessary complexity. As a result, the human approach achieves the same performance with simpler and more cost-efficient cache designs, demonstrating that empirical evaluation remains essential in computer architecture.